

LNA Design Project Final Report

EE 220 RFIC I

San Jose State University

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gm/ID Justification and Device Sizing

The input transistor was implemented using the nmos1v_nat device due to its lower threshold voltage V_{th} of around 150mV, which allows a higher transconductance (g_m) at the same bias condition. Since g_m is proportional to the overdrive voltage V_{ov} , a lower V_{th} enables achieving the desired g_m with a smaller device size, therefore reducing parasitic capacitance and improving high frequency performance.

This advantage is confirmed by the (f_T) vs g_m/I_D analysis in Figure 2. When we size the transistor with a width of 5 μ m, a length of 300nm, and at a g_m/I_D of 6S/A, the nmos1v_nat achieves approximately 13.12GHz compared to 12.32GHz for the nmos1v_lvt. There is a 6.5% improvement in speed.

At a g_m/I_D of 6S/A, nmos1v_nat has a intrinsic gain g_m/g_{ds} of 23.27V/V, while the nmos1v_lvt has a intrinsic gain g_m/g_{ds} of 23.73V/V. Although the intrinsic gain of the nmos1v_nat is slightly lower (~1.9% reduction) than that of the nmos1v_lvt in Figure 3, this tradeoff is acceptable given the significant improvement in speed. Furthermore, due to the higher g_m of nmos1v_nat shown in Figure 1, we get a larger current density making this pick a more power efficient transistor than nmos1v_lvt.

Furthermore, the nmos1v and nmos1v_hvt devices were also evaluated. However, both devices exhibit lower transition frequency (f_T) and reduced current efficiency at the same g_m/I_D operating point as shown in Figures 1-3. Therefore, these devices were not considered in the design.

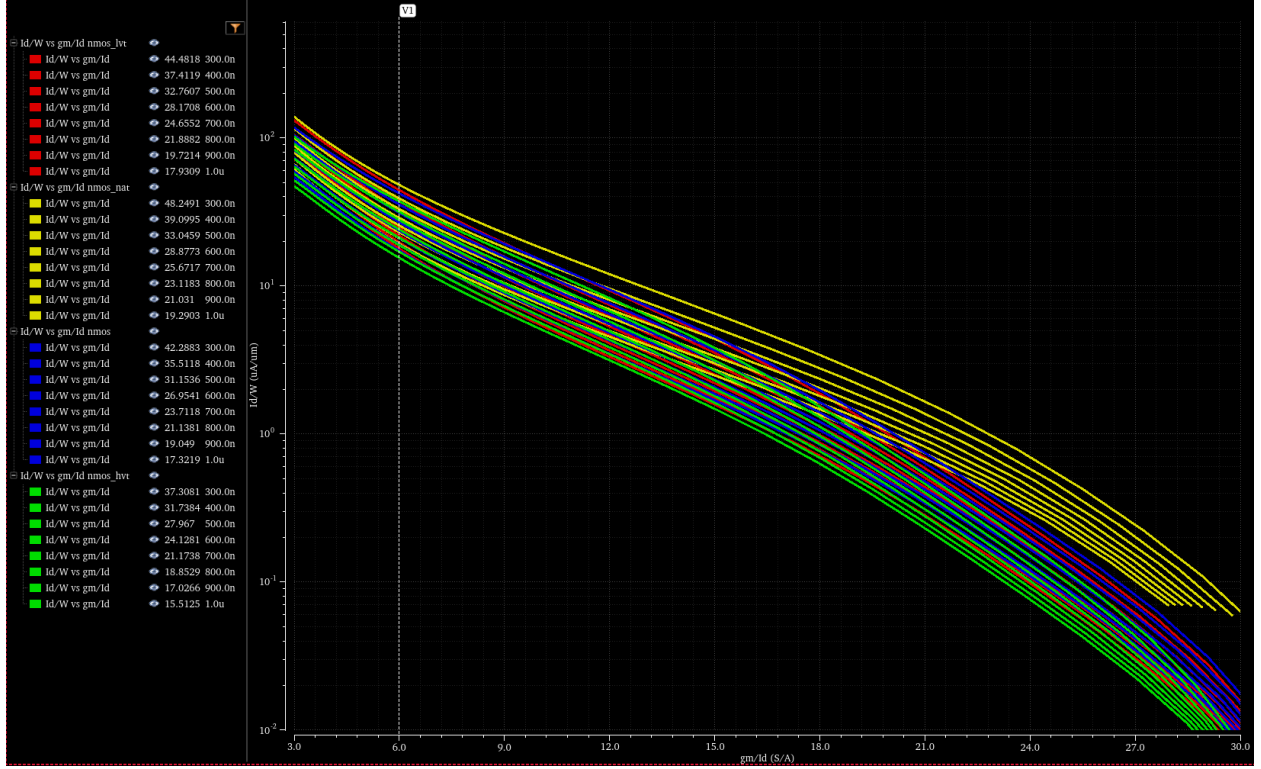


Figure 1: Current density versus g_m/I_d of different types of NMOS.

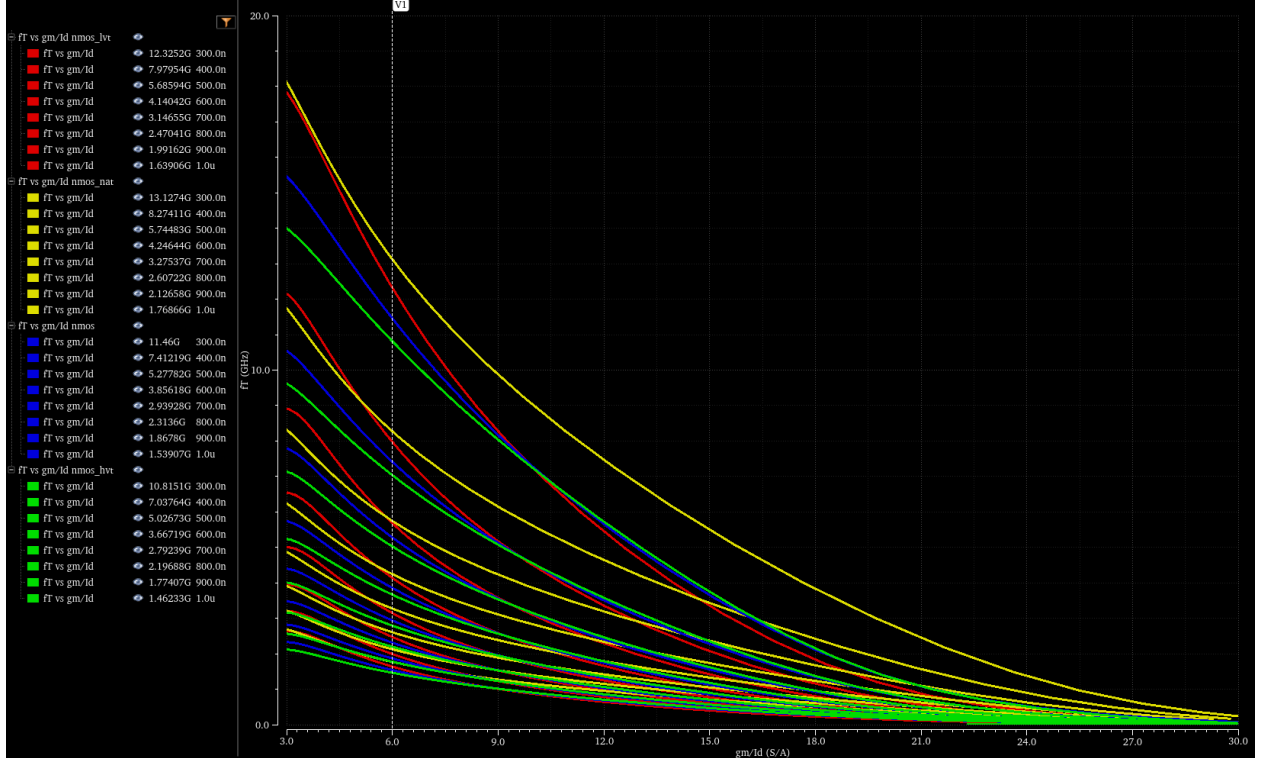


Figure 2: Transition frequency versus g_m/I_d of different types of NMOS.

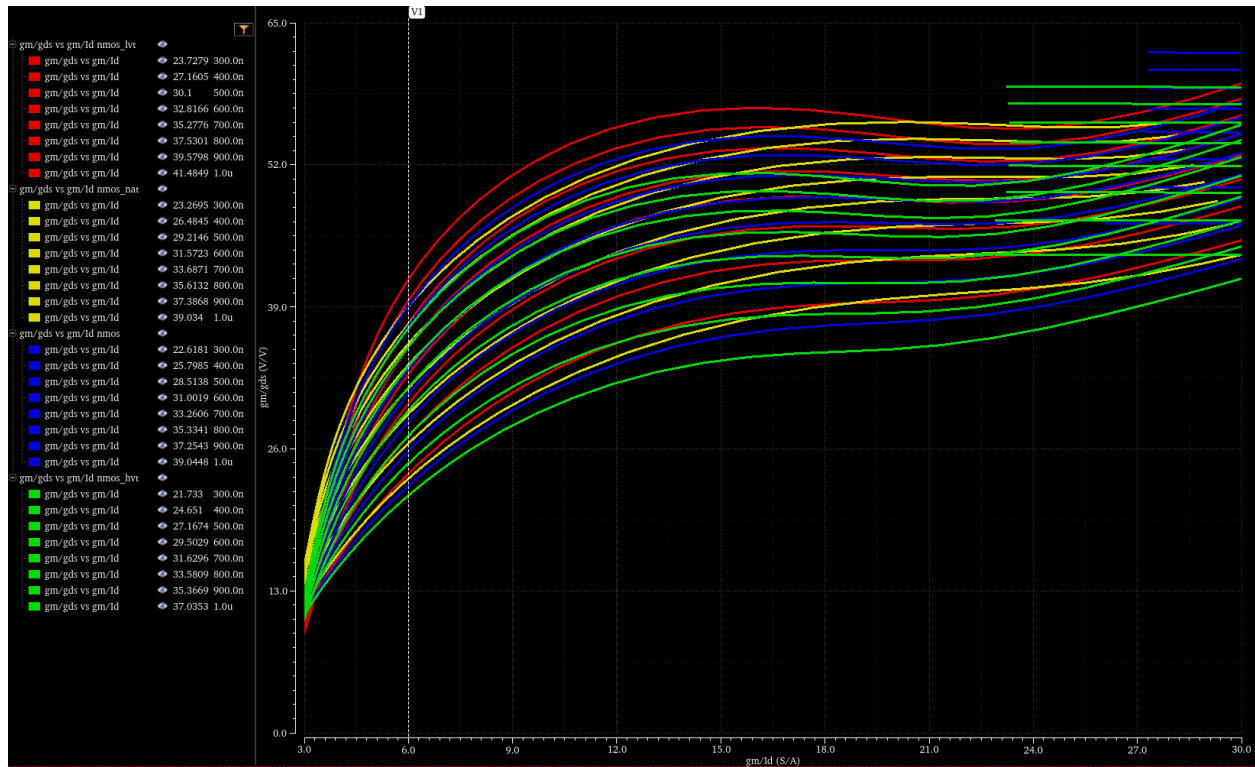


Figure 3: Intrinsic gain versus g_m/I_d of different types of NMOS

Design Specification and approach

LNA Design

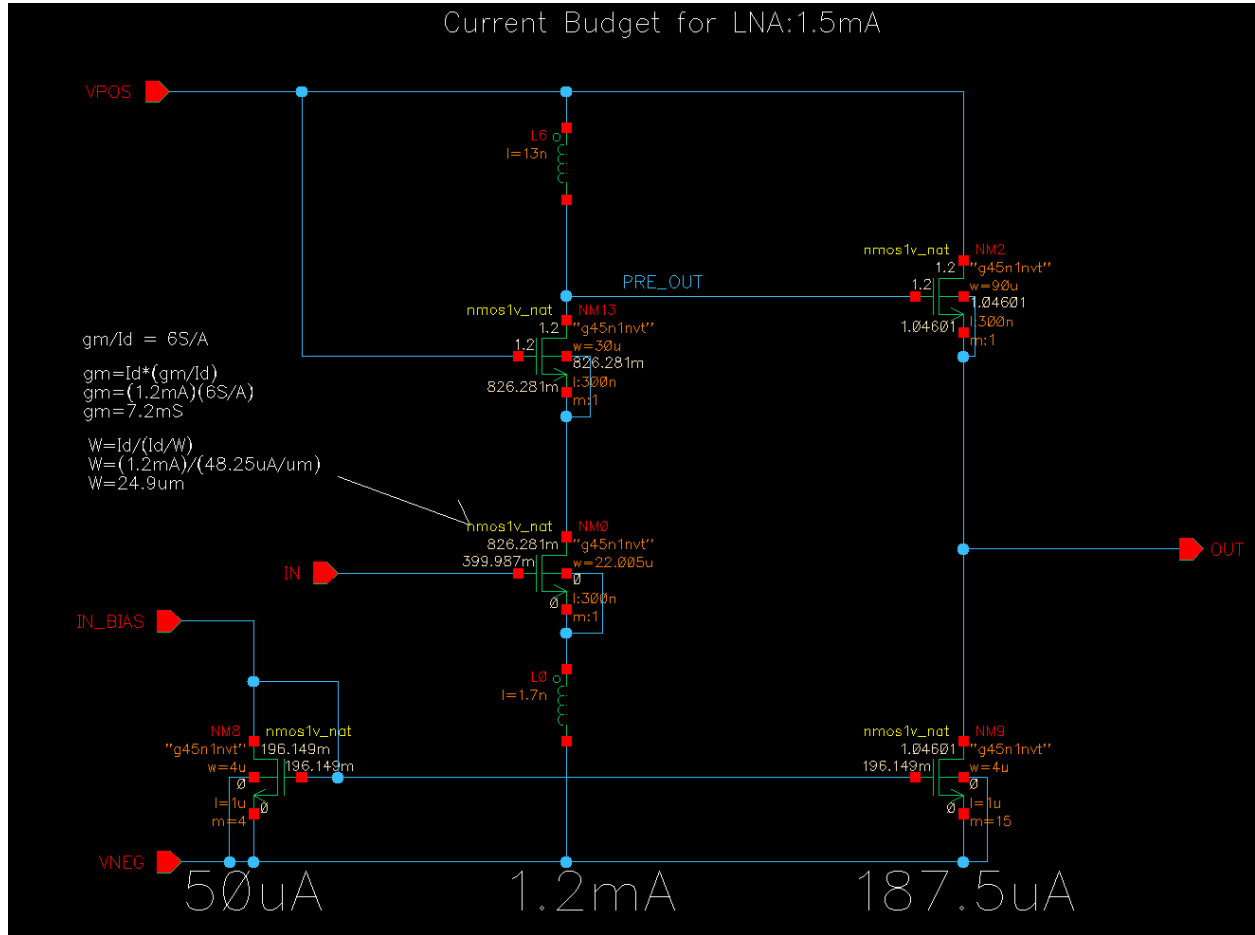


Fig. 4: Schematic of the LNA

The first stage of the LNA uses a common-source amplifier with an inductive source degeneration. The input device, NM0, is a nmos1v_nat which has a V_{th} of 150mV meaning that it will have a higher g_m and can conduct more current at a smaller size. A smaller sized transistor would have smaller amounts of parasitic capacitance which will increase the transient frequency, f_t . Results of these claims can be found in Figures 1-3.

The cascode device, NM13, is used to boost the output impedance looking into the drain by roughly $g_m * r_o^2$. Since the gain of the first stage is based on the transconductance of the input device times the total output impedance, we require a high load impedance so that the design can meet the S21 requirement. A typical PMOS current source would be used, however, it is inefficient since the length of the device needs to be extremely long which would reduce the voltage headroom at "PRE_OUT". As a result, the inductor L6 is used as a load because at DC, this component acts as a short and thus no voltage headroom is used up. As the frequency gets

higher, the inductor starts behaving like a resistive load proportional to the frequency. At 2.45GHz, S21 would be at its maximum before parasitic capacitance shorts “PRE_OUT” to ground. This effect is known as gain peaking. We adjusted the inductor along with the NMOS follower, NM2, until the LNA meets at least 13.5dB.

The second stage of the LNA is a follower implemented with a common-drain NMOS with a current source as a load. The reason for using a follower as the output stage is to provide the output with a very low output impedance which is inversely proportional to the transconductance of the follower. Generating a low output impedance from design instead of using passive components saves die space and reduces any mismatch of the components in the lumped network. A current source is used to keep VGS constant through various voltage swings.

Current Generator Design

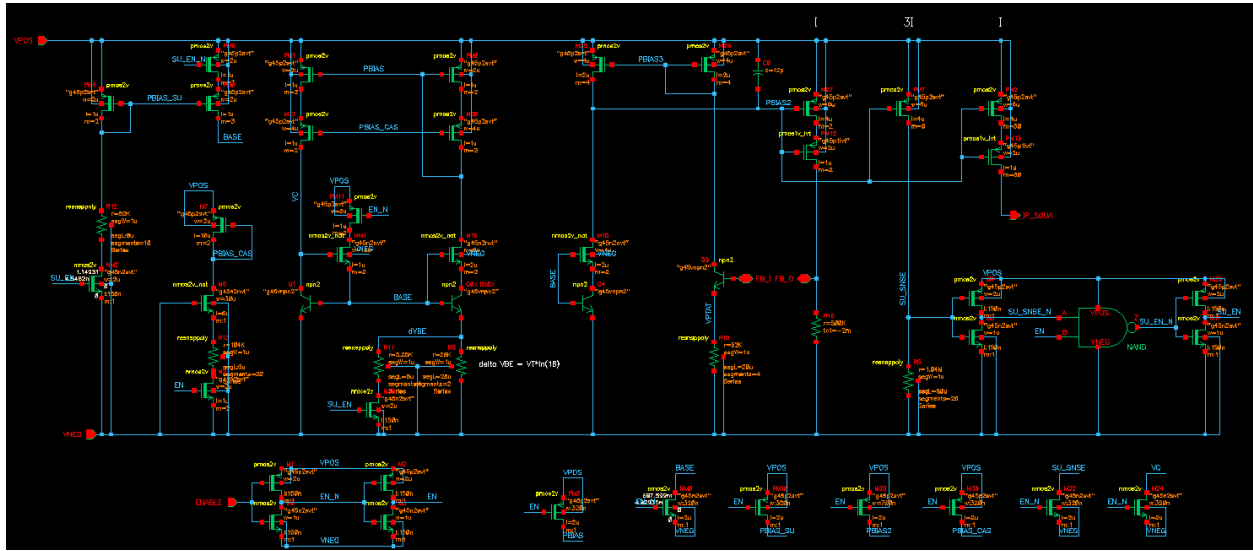


Fig. 5: Schematic of the current generator

The current generator uses the summation of a PTAT and CTAT voltage, and forces the voltage on a resistor to generate a near flat-temperature coefficient output current. The PTAT current is generated by using the difference in current density between NPNs whose ratio equates to 16. The PTAT voltage is then $V_T \ln(16)$ or 70mV which increases with temperature. The current is then mirrored into a resistor which generates a PTAT voltage. Combining this PTAT voltage with a VBE of another NPN which is a CTAT voltage would result in a flat-TC reference voltage which can be forced to another resistor to finally generate the flat-TC output current.

Cadence simulation results

Test Bench

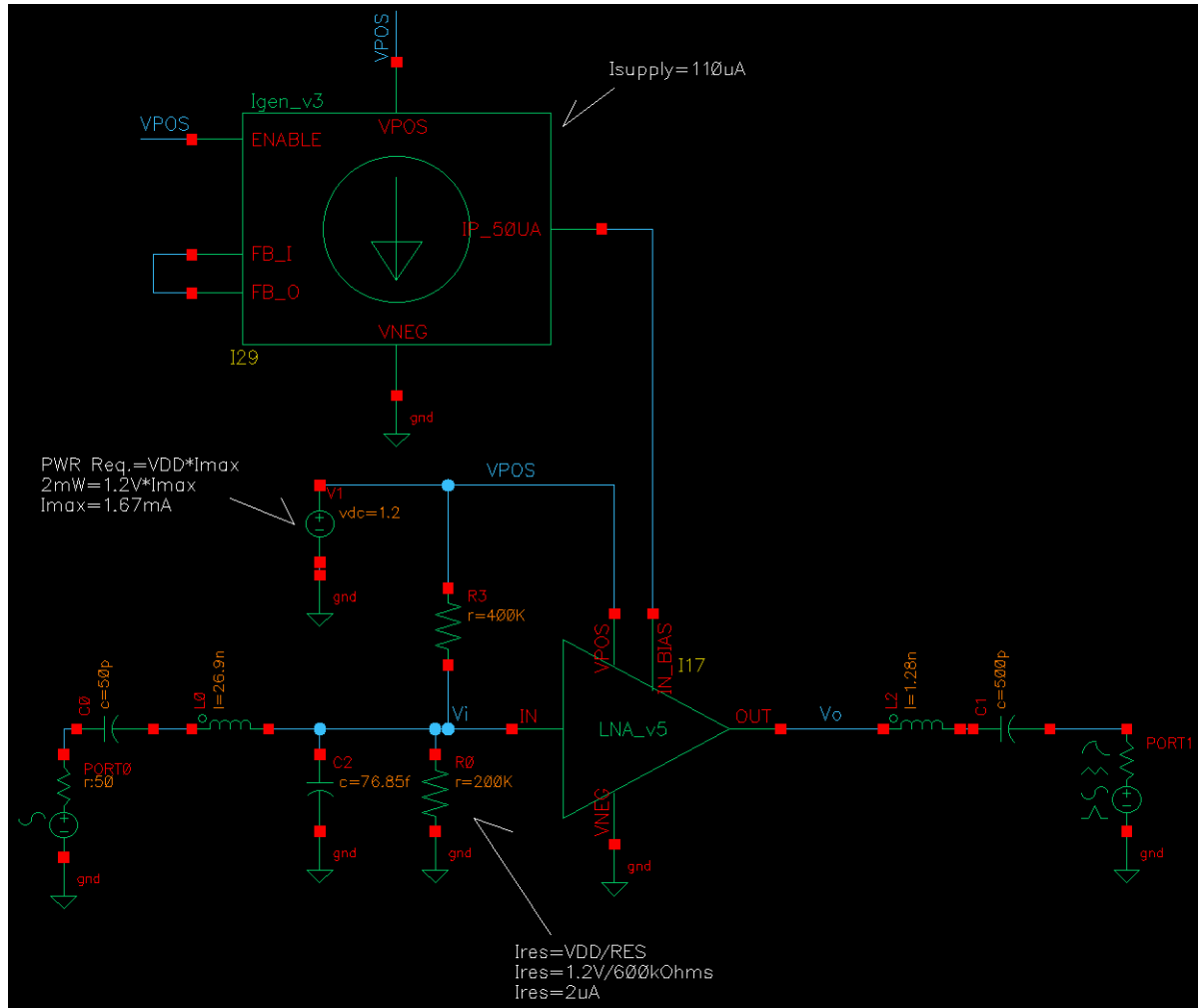


Fig. 6: Testbench of the LNA

LNA biased by an external current source. Matching network both at the input and output. Resistor divider to provide DC bias to the LNA. Decoupling capacitors to block any external DC source.

Simulation results without the matching network



Fig. 7: Simulation result of the LNA without the matching network

Shown above is the simulation result of LNA without the matching network. Here we extracted the input impedance $Z_{M1} = 189.954-j*791.659\Omega$ and the output impedance $Z_{M2} = 49.27-j*19.796\Omega$ at 2.45 GHz.

Simulation results of the matching network

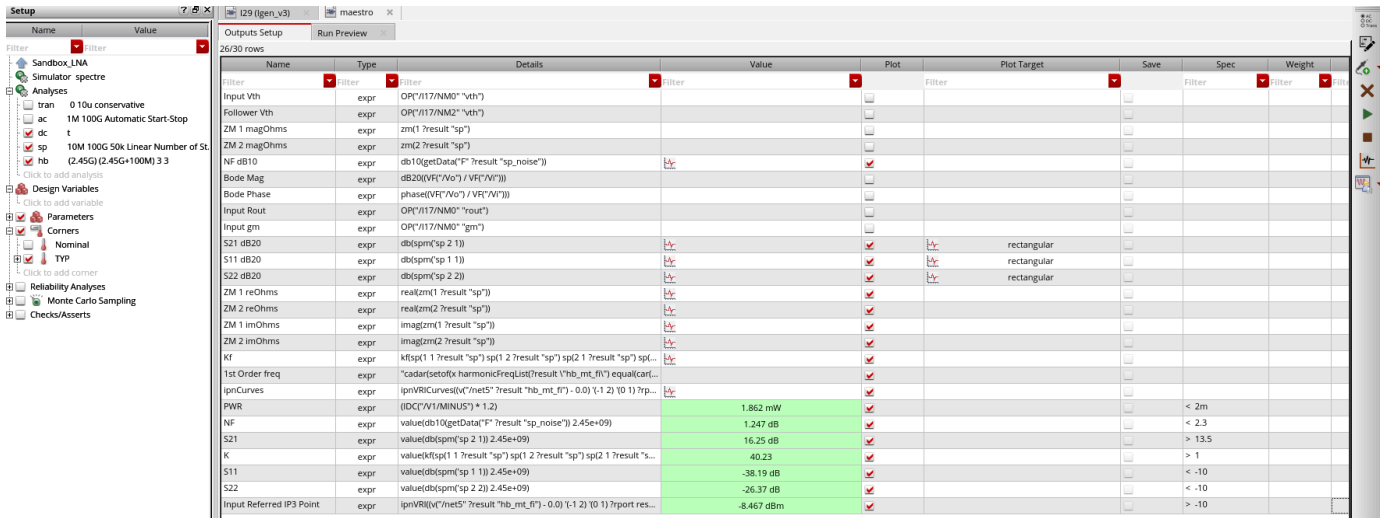


Fig. 8: ADE design variables and output setup

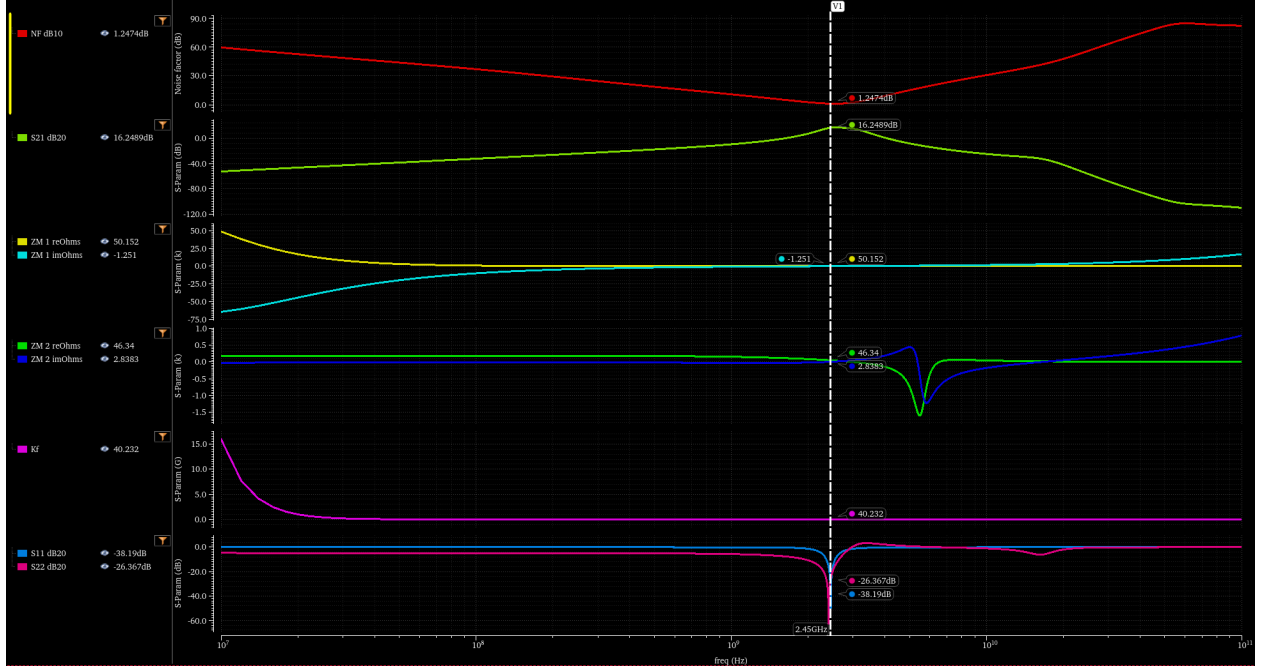


Fig. 9: Simulation result of the LNA with the matching network, including Gain S_{21} , input return loss(S_{11}), output return loss(S_{22}), Noise Figure (NF), Stability (K), input impedance, and output impedance

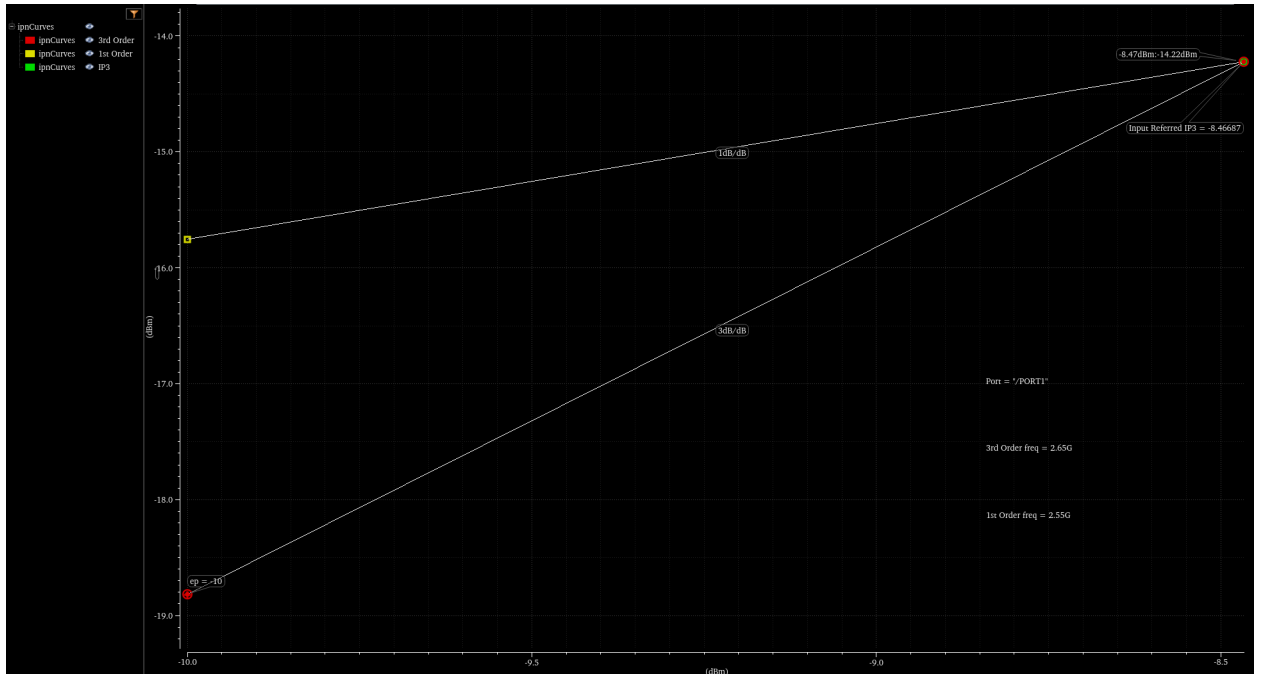


Fig. 10: Input Third-Order Intercept Point (IIP3) of the LNA, showing the extrapolated intersection of the fundamental and third-order intermodulation response.

Parameters	Requirements	Simulation	Satisfied?
Gain S_{21}	$\geq 13.5\text{dB}$	16.25dB	Yes
S_{11}	$\leq -10\text{dB}$	-38.2dB	Yes
S_{22}	$\leq -10\text{dB}$	-26.4dB	Yes
Noise Figure (NF)	$\leq 2.3\text{dB}$	1.25dB	Yes
Power	$\leq 2\text{mW}$	1.86mW	Yes
IIP3	$\geq -10\text{dBm}$	-8.5dBm	Yes
Stability (K)	≥ 1	40.2	Yes

Table 1: Comparison of specified performance requirements and simulated LNA results

The Cadence simulation results confirm that the designed LNA meets all performance requirements for the project. As summarized in the table, the gain S_{21} achieves 16.25dB which exceeds the requirement of 13.5dB. Both input and output return losses, S_{11} and S_{22} are well below -10dB across 2.4-2.5GHz, indicating proper impedance matching at 2.45GHz. The noise figure (NF) is simulated at 1.25dB which is significantly lower than the 2.3dB requirement. This low NF demonstrates that the input device sizing and biasing effectively minimize noise contribution. The power consumption of this circuit is also within the 2mW constraint. Linearity is also satisfied, with an IIP3 of -8.5dBm exceeding the required -10dBm. Last but not least, the stability factor K is consistently greater than 1 (simulated at 40.2), ensuring unconditional stability over 1-10GHz.

Lumped Network Design

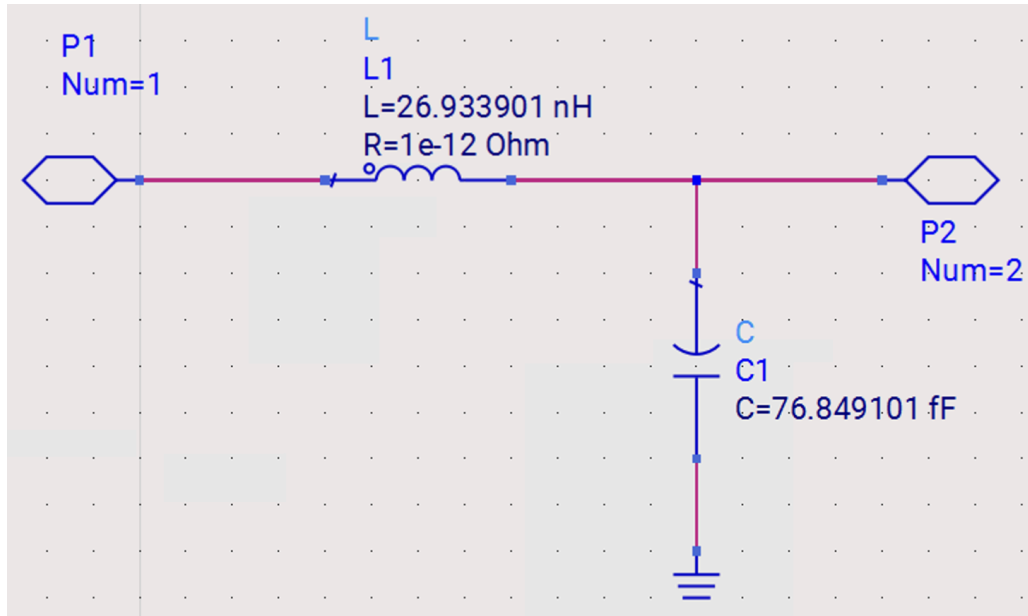


Fig. 11: Input lumped matching network.

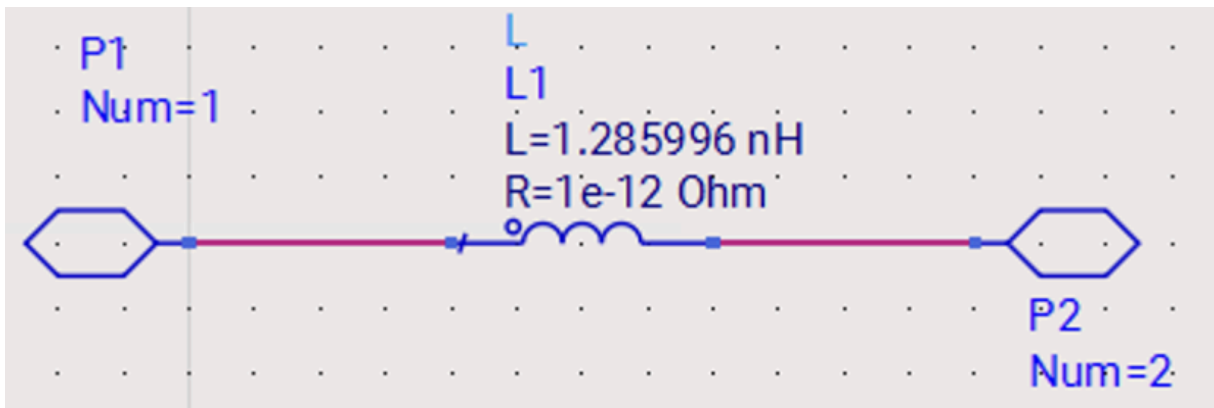


Fig. 12: Output lumped matching network.

The input-side matching network shown in Fig. 11 is a typical low pass L network tuned to be suitable for $Z_{M1} = 189.954 - j*791.659\Omega$ at 2.45 GHz. The initial output impedance value of the LNA was $Z_{M2} = 49.27 - j*19.796\Omega$. Adding a 2-element L network resulted in resonance that degraded results and the real component of the initial impedance was very close to 50Ω , so it was suitable to instead add an inductor in series that had equal and opposite reactance at the characteristic frequency as Fig. 12 shown.

Distributed Network Design

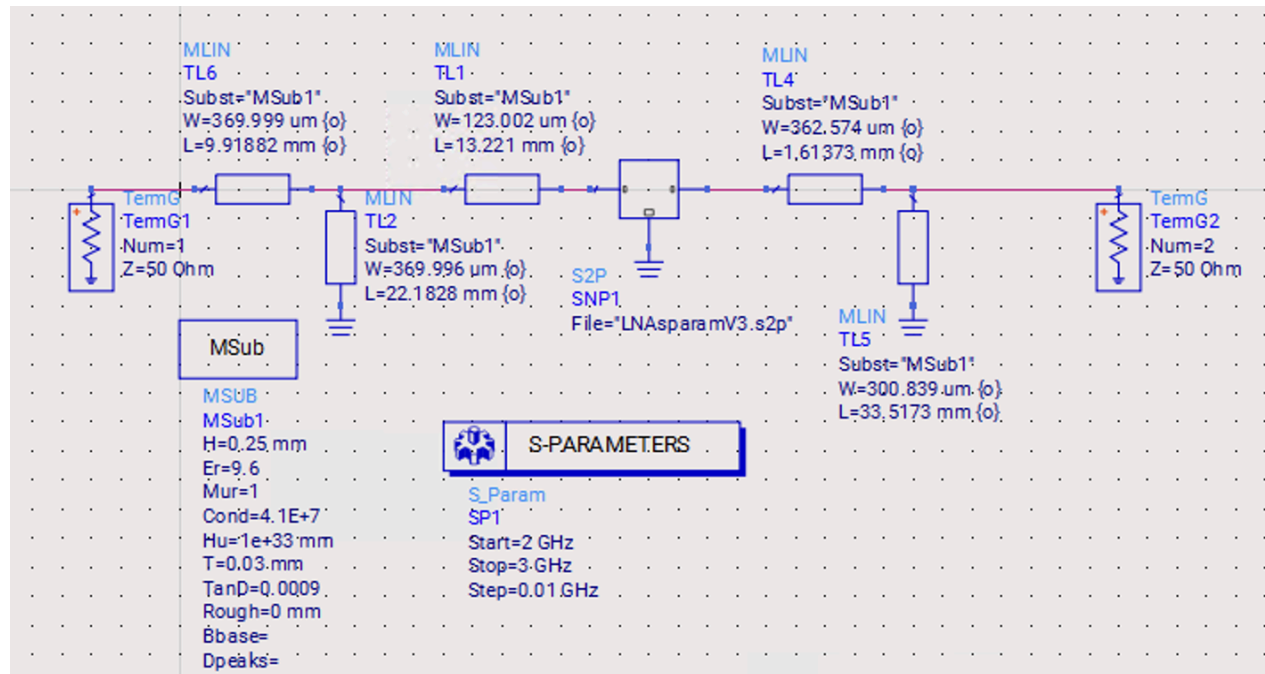


Fig. 13: Distributed network with MLIN.

This architecture gave the best results for the distributed network. I used the ADS optimize tool with various architectures (L input instead of T, no series output MLIN, no shunt output MLIN, etc) to tune the W and L parameters of each MLIN. While the reflection coefficients and VSWR met the requirements, the gain needed to be improved.

Lumped Network Results

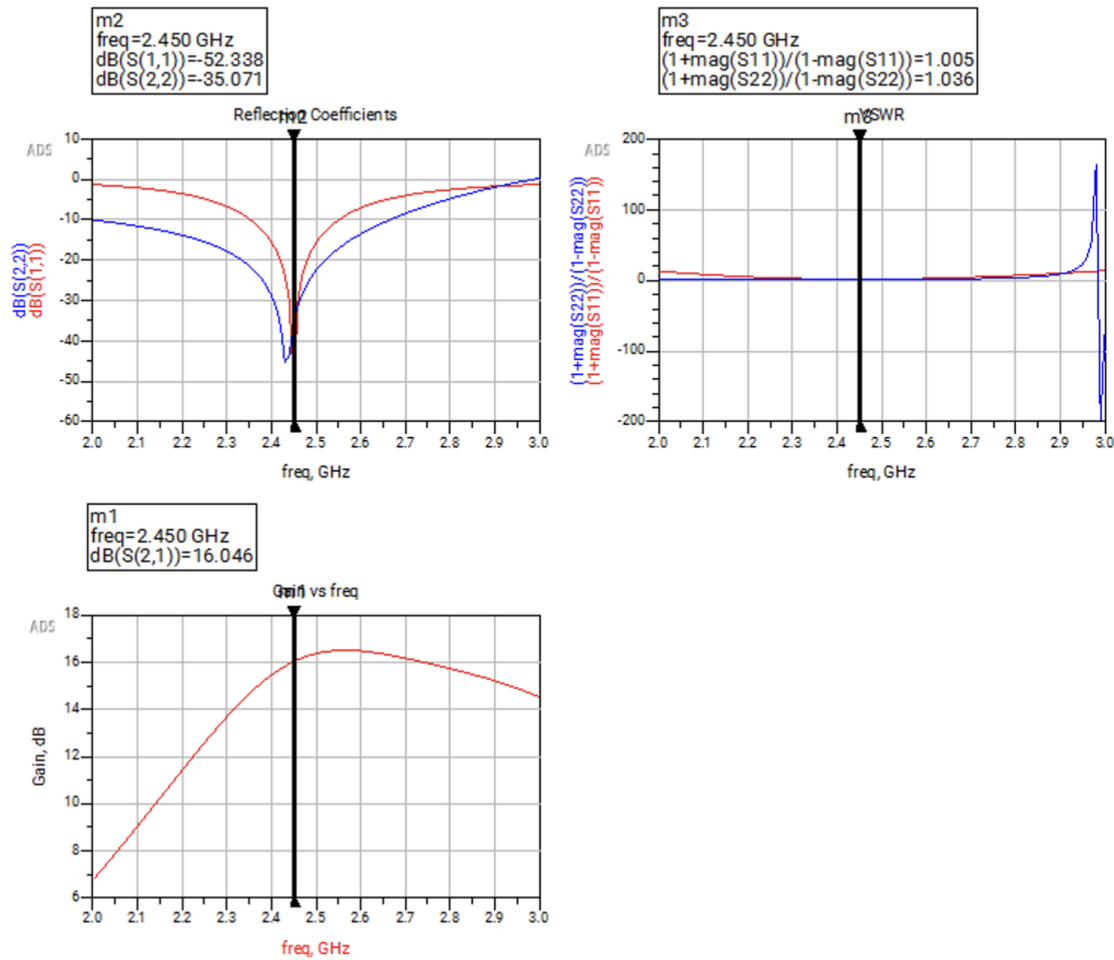


Fig. 14: Lumped network results including input and output return losses, Voltage Wave Standing Ratio (VSWR), and the gain.

The results from the lumped matching network are shown above. The input matching network has been tuned to get the best results. As mentioned earlier, the output matching network was reduced to a single inductor, limiting the tuning that could be done. Nevertheless, the system with the lumped matching network exceeded requirements in reflection coefficients, VSWR requirements, and gain.

Distributed Network Results

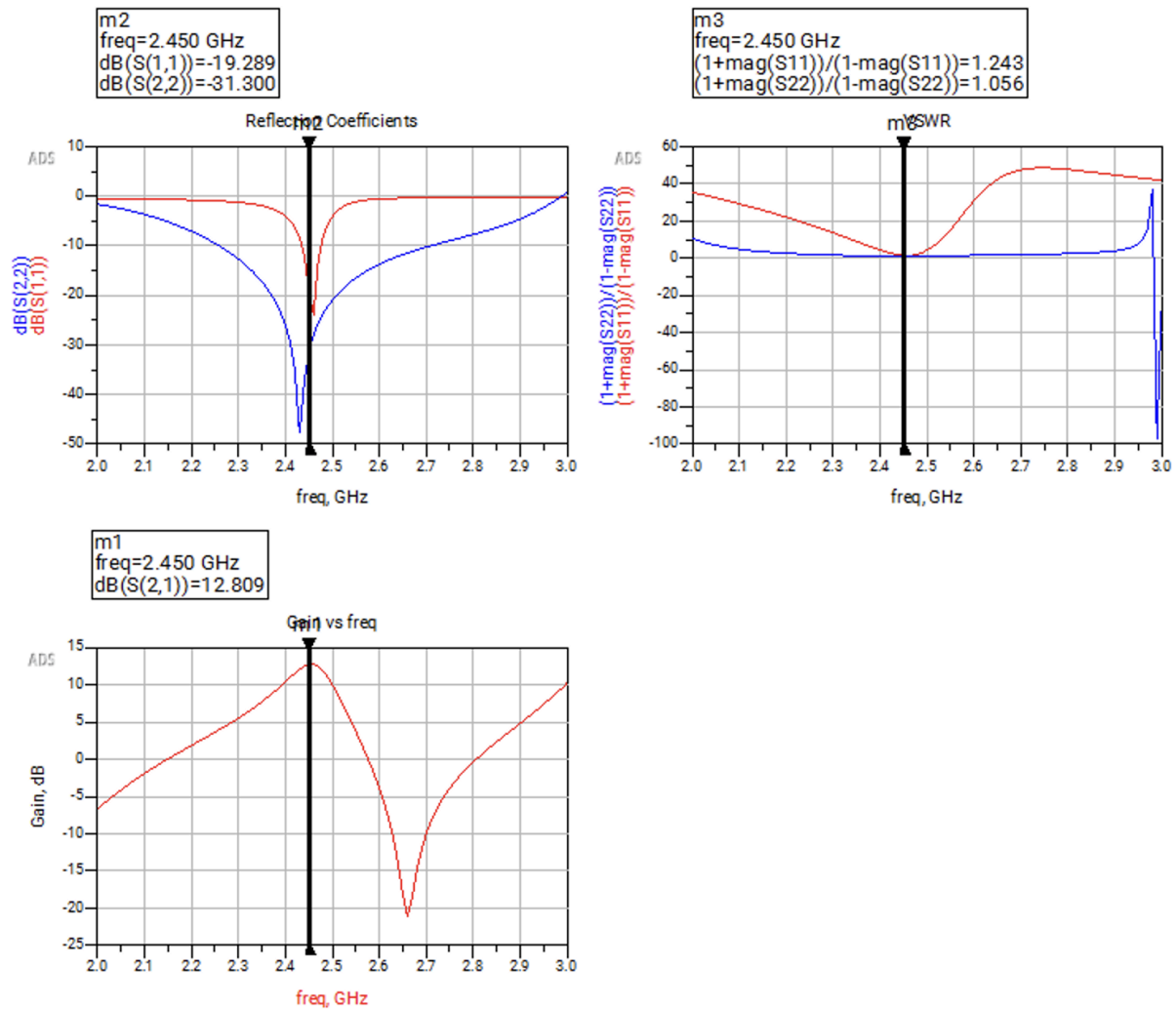


Fig. 15: Distributed network results including input and output return losses, Voltage Wave Stand Ratio (VSWR), and the gain.

While the distributed matching network led to satisfactory performance in reflection coefficients and VSWR, the gain did not meet requirements. Once again, there are signs of resonance near the band of interest, which explains the degraded performance compared to the results with the lumped network. Potential methods to improve results include redesigning the input network and modifying the LNA itself.

Tradeoffs Between Lumped and Distributed Networks

Due to the resonance in the distributed network, it performed worse than the lumped network in all metrics:

Network Type	Return Loss	Bandwidth	Frequency Sensitivity
Lumped	-52.338 dB	0.8 GHz	Moderate
Distributed	-19.289 dB	0.1 GHz	High

The lumped system performed better in the three metrics mentioned above in addition to being easier to design (without layout). However, this data should not be used to form general conclusions about comparisons between lumped networks and distributed networks, as resonance is not likely to be a factor in most cases. Generally speaking, a distributed system should have better bandwidth and will consume less chip area.